

MARBLE BAR, AustraliaWMO: **953170**Lat: **21.176S**Lon: **119.750E**Elev: **183**StdP: **99.15**Time Zone: **8.00 (E08)**Period: **01-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.2	11.7	-13.6	1.2	28.8	-10.5	1.6	26.3	9.1	19.5	8.4	20.7	1.4	140	0.450

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	13.9	44.1	19.9	42.9	20.0	41.9	20.1	26.1	30.5	25.5	30.5	25.1	30.5	4.0	130

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
25.2	20.9	27.1	24.5	20.0	27.2	23.9	19.2	27.5	82.2	30.1	79.9	30.1	78.1	30.3	35.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 10.3	8.4	7.4	DB	7.1	46.1	0.9	1.5	6.4	47.1	5.9	48.0	5.3	48.8	4.7	49.8	(4)
(5)			WB	3.0	28.5	1.2	2.9	2.1	30.5	1.4	32.2	0.7	33.8	-0.2	35.9	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	27.9	33.3	32.5	31.4	29.1	24.0	20.3	20.1	21.7	25.6	30.4	32.4	33.9	(6)	
	DBStd	5.68	2.99	2.95	2.88	2.33	2.69	2.38	2.38	2.24	2.44	2.81	2.45	2.30	(7)	
	HDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0	(8)	
	HDD18.3	21	0	0	0	0	1	8	10	2	0	0	0	0	(9)	
	CDD10.0	6524	723	631	665	574	434	311	313	361	467	631	673	742	(10)	
	CDD18.3	3503	464	398	407	324	177	68	64	105	217	373	423	484	(11)	
	CDH23.3	50537	7124	6029	5949	4321	1807	574	663	1277	2870	5597	6608	7718	(12)	
	CDH26.7	31547	4779	3926	3723	2488	799	138	179	494	1584	3622	4473	5341	(13)	
(14) Wind	WSAvg	3.3	3.7	3.4	3.2	2.9	3.1	3.2	2.9	3.0	3.0	3.5	3.7	3.8	(14)	
(15) Precipitation	PrecAvg	388	80	111	59	20	23	21	11	8	1	2	10	46	(15)	
	PrecMax	798	204	347	277	143	187	165	108	89	18	18	71	315	(16)	
	PrecMin	169	2	10	1	0	0	0	0	0	0	0	0	0	(17)	
	PrecStd	156	54	87	69	35	37	35	22	18	3	4	15	57	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	45.3	45.1	42.7	40.2	35.7	31.5	31.7	34.4	38.4	42.0	43.7	45.7	(19)	
		MCWB	20.2	20.1	20.9	19.6	17.0	17.3	15.3	15.7	16.7	18.5	18.7	19.9	(20)	
	2%	DB	43.8	43.4	41.1	38.6	34.3	29.7	30.2	32.7	36.9	40.9	42.5	43.8	(21)	
		MCWB	20.3	20.4	20.7	19.1	16.8	16.1	15.3	15.2	16.3	18.1	18.7	20.0	(22)	
	5%	DB	42.4	41.6	39.9	37.2	32.7	28.3	28.7	31.2	35.6	39.8	41.5	42.5	(23)	
		MCWB	20.9	21.0	20.6	18.8	16.7	15.3	14.8	14.5	16.1	18.0	18.7	20.0	(24)	
	10%	DB	41.0	39.8	38.4	35.8	31.2	26.8	27.1	29.6	34.2	38.5	40.1	41.2	(25)	
		MCWB	21.5	21.7	20.9	18.8	16.3	14.5	14.0	14.4	15.9	17.8	18.7	20.3	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	26.5	26.6	26.4	24.9	22.5	23.1	21.0	22.5	27.6	22.7	24.6	25.7	(27)	
		MCDB	31.5	31.3	30.1	28.5	27.3	24.6	23.9	23.3	28.2	29.9	31.6	31.2	(28)	
	2%	WB	25.9	25.8	25.6	23.8	20.7	19.7	18.7	17.6	19.1	21.3	23.1	24.8	(29)	
		MCDB	30.9	30.6	29.7	28.9	26.9	24.3	24.9	26.9	28.4	31.4	31.3	31.5	(30)	
	5%	WB	25.5	25.4	25.0	22.8	19.6	18.6	17.5	16.4	17.9	20.4	22.0	24.2	(31)	
		MCDB	30.9	30.6	29.6	29.3	26.4	24.0	24.7	27.3	28.9	32.2	32.6	31.7	(32)	
	10%	WB	25.0	25.0	24.5	22.0	18.3	17.2	15.9	15.3	17.0	19.6	21.2	23.6	(33)	
		MCDB	30.9	30.7	29.8	29.5	26.8	23.2	24.4	27.2	30.3	32.5	32.3	32.4	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	12.7	12.2	11.8	12.9	12.9	12.7	13.9	15.6	16.3	15.3	14.6	13.9	(35)	
		MCDBR	15.1	15.0	14.1	14.4	15.2	14.4	16.0	18.1	18.2	16.0	15.5	15.5	(36)	
	5% WB	MCWBR	4.1	4.0	3.6	3.8	4.9	5.2	6.1	6.6	5.9	4.4	4.1	3.8	(37)	
		MCWBR	11.7	11.5	10.7	11.0	10.6	10.5	12.7	14.7	15.8	14.7	14.4	13.1	(38)	
(40) Clear-Sky Solar Irradiance	taub		0.394	0.386	0.375	0.353	0.334	0.319	0.320	0.321	0.340	0.365	0.373	0.375	(40)	
		taud	2.468	2.508	2.517	2.519	2.491	2.525	2.495	2.447	2.413	2.381	2.403	2.477	(41)	
	Ebn at Noon		951	946	931	912	891	888	897	931	951	955	965	969	(42)	
		Edh at Noon	119	113	107	99	94	87	93	105	117	127	127	118	(43)	
(44) All-Sky Solar Radiation	RadAvg		7.35	7.02	6.36	5.70	4.81	4.36	4.72	5.84	6.84	7.59	8.04	7.95	(44)	
	RadStd		0.51	0.56	0.50	0.28	0.37	0.34	0.30	0.18	0.25	0.26	0.28	0.39	(45)	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (2 neighbors)		+0.49	N/A	N/A	N/A	+0.48	+0.52	N/A	N/A	+158

Nomenclature: See separate page